

Eigenvalues and inverse problems

The sharp generic threshold for spectral-subspace rotation

Difficulty: challenging **Importance:** interesting to the community

Status: explicit open constant problem; no later resolution located

Let A be a self-adjoint, possibly unbounded operator on a separable complex Hilbert space. Suppose $\sigma(A) = \sigma \cup \Sigma$, where the two nonempty closed sets satisfy $d = \text{dist}(\sigma, \Sigma) > 0$. For a bounded self-adjoint perturbation V , set

$$P = E_A(\sigma), \quad Q = E_{A+V}(O_{d/2}(\sigma)), \quad O_r(S) = \{x \in \mathbb{R} : \text{dist}(x, S) < r\},$$

where E_B denotes the spectral projection measure of B .

Does $\|V\| < d/2$ always imply $\|P - Q\| < 1$? All norms are operator norms. Equivalently, is the optimal universal constant $c_{\text{opt}} = 1/2$ for guaranteeing that the maximal angle $\arcsin \|P - Q\|$ is strictly below $\pi/2$?

The two spectral sets may interlace: no ordering or disjoint-convex-hull hypothesis is imposed. The source's operator formulation is retained; Hermitian matrix invariant-subspace perturbation is its finite-dimensional setting. This problem concerns subspace orientation after perturbation, beyond preservation of a spectral gap.

References

A. Seelmann, *Notes on the subspace perturbation problem for off-diagonal perturbations*, Proceedings AMS 144 (2016), 3825–3832, introduction p.3826, the paragraph on general perturbations ([primary manuscript; journal](#)). Seelmann, *On an estimate in the subspace perturbation problem*, Journal d'Analyse Mathématique 135 (2018), 313–343 ([primary manuscript](#)). Seelmann, *Unifying the treatment of indefinite and semidefinite perturbations in the subspace perturbation problem*, Operators and Matrices 15 (2021), 1181–1188, Theorems 1.1–1.2 ([primary paper](#)).

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The 2021 generic theorem retains the sufficient constant $c_{\text{crit}} = 0.4548399 \dots$; its sharp theorem at a larger threshold assumes extra spectral separation. The 2018 paper's solved optimization problem determines an upper angle bound, not the conjectured threshold. Searches for “subspace perturbation”, “optimal constant”, “1/2”, “Seelmann”, “conjecture”, and 2024–2026 found no resolution. No recent explicit reaffirmation of the exact endpoint conjecture was located; the status evidence is bounded.